



Maneuvering

Maneuvering a Cave Problem

Problem:

The task is to count all the possible paths from top left to bottom right of a $m \times n$ matrix with the constraints that from each cell you can either move only to right or down.

Examples :

Input : $m = 2, n = 2;$

Output : 2

There are two paths

$(0, 0) \rightarrow (0, 1) \rightarrow (1, 1)$

$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1)$

Let $\text{NumberOfPaths}(m, n)$ be the count of paths to reach row number m and column number n in the matrix, $\text{NumberOfPaths}(m, n)$ can be recursively written as following.

```
class Main {  
    static int numberOfPaths(int m, int n){  
        if (m == 1 || n == 1)  
            return 1;  
        return numberOfPaths(m-1,n)+numberOfPaths(m,n-1);  
        // + numberOfPaths(m-1, n-1);  
    }  
  
    public static void main(String args[])  
    {  
        System.out.println(numberOfPaths(3, 3));  
    }  
}
```



THANK YOU